CSL202 : Data Structures
2025-26-M
HOME ASSIGNMENT - I

Due Date: 23:59 , Aug 31, 2025.

Instructions:

1. Submissions will not be accepted after the due date. Make sure to finish and upload your work on time.
2. Each problem must be solved in a separate C (or CPP etc.) program file.
3. Name the files in the following format:
    - pro_1.c → Problem 1
    - pro_2.c → Problem 2 and so on.
4. Put all 10 .c files in a single folder and make it zip file with name as:
    - Name_RollNumber.zip
    - Example- RAMESH_B2412345.zip
5. Upload the .zip file to the assignment in the CANVAS.

Q1. Circular Queue Rotation

Write a function to rotate a circular queue by k positions to the right.

Example:

Input: Queue = [1,2,3,4,5], k=2
Output: [4,5,1,2,3]

Q2. Implement a Queue using Two Stacks

Write a program to implement a queue using two stacks. Support both enqueue and dequeue operations.

Example:

Input: enqueue(1), enqueue(2), enqueue(3), dequeue()
Output: 1

Q3. Queue Simulation – Customer Service

Simulate a queue of customers at a service desk:

- Each customer has a service time t.
- Serve customers in order of arrival.
- Print the total waiting time for all customers.

Example:

Input: serviceTimes = [2, 3, 1]
Output: Total Waiting Time = 5
Explanation: waiting time = 0 + 2 + (2+3) = 7

Q4. Reverse the First k Elements of a Queue

Write a function to reverse the first k elements of a queue while keeping the other elements in the same relative order.

Example:

Input: Queue = [1, 2, 3, 4, 5], k = 3
Output: [3, 2, 1, 4, 5]

Q5. Implement a Circular Queue using Arrays

Write a program to implement a circular queue using arrays. Support all standard operations:

- enqueue(x)
- dequeue()
- peek()
- isFull()
- isEmpty()

Example:

Input: enqueue(1), enqueue(2), enqueue(3), dequeue(), enqueue(4), peek()
Output: 2

Q6. Check if a Queue is a Palindrome

Write a function to check whether a queue of integers forms a palindrome.

Example:

Input: [1, 2, 3, 2, 1]
Output: True

Input: [1, 2, 3, 4]
Output: False

Q7. Dota2 Senate

The Senate consists of senators from two parties: Radiant and Dire. The senators are given in a string s where each character is either 'R' (Radiant) or 'D' (Dire).
Each round:

- A senator bans one member of the other party.
- The banned senator cannot participate in future rounds.
- The game continues until all senators are from one party. Return the winning party ("Radiant" or "Dire").

Example:

Input: "RD"
Output: "Radiant"

Input: "RDD"
Output: "Dire"

Q8. Number of Students Unable to Eat Lunch

The school cafeteria has a queue of students and a stack of sandwiches.

Students are in a queue represented by array students (1 = circular sandwich, 0 = square sandwich).

Sandwiches are in a stack represented by array sandwiches.

If the student at the front prefers the top sandwich, they take it and leave. Otherwise, they move to the end of the queue.
Return the number of students unable to eat lunch.

Example:

Input: students = [1,1,0,0], sandwiches = [0,1,0,1]
Output: 0

Input: students = [1,1,1,0,0,1], sandwiches = [1,0,0,0,1,1]
Output: 3

Q9. First Unique Character in a String

Given a string s, find the first non-repeating character in it and return its index. If it does not exist, return -1. Use a queue to help in the process.

Example:

Input: "swiss"
Output: 0

Input: "programming"
Output: 0

Input: "aabbcc"
Output: -1

Q10. Number of Recent Calls

You have a RecentCounter class that counts the number of recent requests in the last 3000 milliseconds.

Implement ping(t) that records a new request at time t (milliseconds).

Return the number of requests made from [t - 3000, t].

Example:

Input: ping(1), ping(100), ping(3001), ping(3002)
Output: 1, 2, 3, 3

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